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MATHEMATICS

RATIO AND PROPORTIONAL REASONING

Name: _____

Date: _____

Refine Judgement and Communication in Proportional Problems

Step 12

Elementary (Grades 2–4)

Today I am learning to refine my judgement and clearly communicate my thinking when solving proportional problems.

1. Check the Reasonableness

Decide if each answer is reasonable.
Tick ✓ Reasonable or ✗ Not Reasonable.

	Question	Answer	Reasonable? ✓ or ✗
a)	5 apples cost $4.50. How much for 12 apples?	$18.00	☐ / ☐
b)	A recipe uses 2 cups of rice to feed 4 people. How much rice for 10 people?	10 cups	☐ / ☐
c)	A car uses 6 litres of fuel to travel 90 km. How far can it travel using 18 litres?	270 km	☐ / ☐
d)	8 pencils cost $3.20. How much for 20 pencils?	$12.80	☐ / ☐

Choose one that is NOT reasonable.
Explain why. _____

2. Explain Your Thinking

Solve each problem. Show your working and explain your thinking.

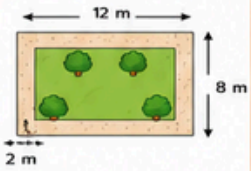
a) 3 packets of biscuits cost $6.00.
How much do 7 packets cost?

Working:

Answer: $_____

Explain your thinking: _____

b) A garden is 12 m long and 8 m wide.
A path 2 m wide goes all around the outside.
What is the area of the path?



Working:

Answer: _____ m²

Explain your thinking: _____

3. Compare and Justify

Two students solved the same problem.
15 apples cost $6.00. How much do 25 apples cost?

Sam's Solution

Unit price = $6.00 ÷ 15
= $0.40 per apple
25 apples cost
$0.40 × 25 = $10.00

Tia's Solution

25 is 10 more than 15.
10 apples cost $6.00 ÷ 15 × 10
= $4.00
So 25 apples cost
$6.00 + $4.00 = $10.00

a) What is the answer? $_____

b) Do you agree with both solutions? Yes ☐ No ☐

c) Explain why their methods are the same or different.

4. Open-Ended Problem

A café sells smoothies.

- 2 cups of berries make 3 smoothies.
- 1 banana is needed for every 2 smoothies.
- 1 cup of yoghurt is needed for every 3 smoothies.

a) If the café wants to make 18 smoothies, how many cups of berries, bananas and cups of yoghurt are needed?
Show your working.

Ingredient	Ratio	For 18 smoothies
Berries (cups)	2 : 3	_____
Bananas	1 : 2	_____
Yoghurt (cups)	1 : 3	_____

b) If the café only has 8 bananas, how many smoothies can they make?
Show your working.

5. Communicate Mathematically

A class made scale drawings of a rectangle.
The real rectangle is 8 m long and 6 m wide.
Here are two scale drawings.

a) Which drawing uses a larger scale? _____

b) What scale was used for each drawing?
Drawing A: 1 cm represents _____ m
Drawing B: 1 cm represents _____ m

c) Explain how you worked this out.

6. Real-Life Decision

You need to buy dog food.
Two bags are on sale.

Bag A	Bag B
4 kg for $28.00	10 kg for $62.00

a) What is the unit price (price per kg) for each bag?
Bag A: $_____ per kg Bag B: $_____ per kg

b) Which bag is the better value? _____

c) Write a sentence to explain your choice.

7. Challenge Problem

A recipe for lemonade uses 3 parts lemon juice to 5 parts water.

a) If you use 600 mL of lemon juice, how much water do you need?
Show your working.

_____ mL

b) If you use 2.4 litres of water, how much lemon juice do you need?
Show your working.

_____ mL

8. Reflect

Read each statement. Tick the level that best describes you.

I can...	Not yet	Sometimes	Mostly	Always
use proportional reasoning to solve problems.				
judge if answers are reasonable.				
explain my thinking clearly.				
choose efficient strategies.				
communicate my solutions using words, numbers and models.				

Think!
Clear thinking and good communication make a great mathematician!

I can...

- ☐ check if answers are reasonable.
- ☐ explain my thinking.
- ☐ compare different methods.
- ☐ solve problems in real-life situations.
- ☐ communicate clearly.

Key Vocabulary

- **proportional** – having a constant ratio between quantities
- **scale** – a ratio that shows how a drawing compares to real life
- **rate** – a comparison of two quantities
- **unit rate** – the amount per one unit
- **efficient** – using the best or least amount of resources
- **justify** – give reasons or explain why

Great job!

You are refining your judgement and communicating like a confident mathematician! Keep up the fantastic work!